

Math 142, S12
Exam 1

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	20	
3	20	
4	18	
5	24	
Total	100	

Good Luck !

Problem 1. (18 points) Evaluate the following integrals.

(a) $\int e^x \sqrt{1+e^x} dx$

$$\begin{aligned} \frac{u=1+e^x}{du=e^x dx} \int \sqrt{u} du \\ &= \frac{2}{3} u^{3/2} + C \\ &= \frac{2}{3} (1+e^x)^{3/2} + C \end{aligned}$$

(b) $\int \frac{x}{1+x^4} dx$

$$\begin{aligned} \frac{u=x^2}{du=2x dx} \int \frac{1}{2} \frac{1}{1+u^2} du \\ &= \frac{1}{2} \arctan(u) + C \\ &= \frac{1}{2} \arctan(x^2) + C \end{aligned}$$

(c) $\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta d\theta$

$$\begin{aligned} &= \int_0^{\pi/4} \tan^2 \theta \sec^2 \theta \sec^2 \theta d\theta \\ &= \int_0^{\pi/4} \tan^2 \theta (1+\tan^2 \theta) \sec^2 \theta d\theta \\ \frac{u=\tan \theta}{du=\sec^2 \theta} \int_0^1 \overset{\rightarrow \tan(\pi/4)}{u^2(1+u^2)} du &= \int_0^1 (u^2+u^4) du \\ &= \left(\frac{u^3}{3} + \frac{u^5}{5} \right) \Big|_0^1 \\ &= \left(\frac{1}{3} + \frac{1}{5} \right) - (0) = \frac{8}{15} \end{aligned}$$

Problem 2. (20 points) Evaluate the following integrals.

(a) $\int x e^{-3x} dx$

$$u = x, dv = e^{-3x} dx$$

$$du = dx, v = -\frac{e^{-3x}}{3}$$

$$\int x e^{-3x} dx = x \cdot \left(-\frac{e^{-3x}}{3}\right) - \int \left(-\frac{e^{-3x}}{3}\right) dx$$

$$= -\frac{x e^{-3x}}{3} - \frac{e^{-3x}}{9} + C$$

$$= -e^{-3x} \left(\frac{x}{3} + \frac{1}{9}\right) + C$$

(b) $\int \sin \sqrt{t} dt$

Let $x = \sqrt{t}$, then $dx = \frac{1}{2\sqrt{t}} dt$ or $dt = 2x dx$

$$\int \sin \sqrt{t} dt \xrightarrow{x=\sqrt{t}} \int \sin x \cdot (2x dx)$$

$$= \int 2x \sin x dx$$

$$\text{Let } u = 2x, dv = \sin x dx$$

$$du = 2dx, v = -\cos x$$

$$\text{then } \int 2x \sin x dx = 2x(-\cos x) - \int (-\cos x) 2 dx$$

$$= -2x \cos x + 2 \sin x + C$$

thus

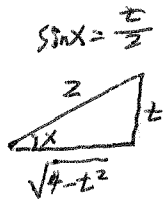
$$\int \sin \sqrt{t} dt = -2\sqrt{t} \cos \sqrt{t} + 2 \sin \sqrt{t} + C$$

Problem 3. (20 points) Evaluate the following integrals.

(a) $\int \sqrt{4-t^2} dt$

let $t = 2 \sin X$, then $dt = 2 \cos X dX$

$$\int \sqrt{4-t^2} dt \stackrel{t=2\sin X}{=} \int \sqrt{4-(2\sin X)^2} \cdot 2 \cos X dX$$



$$= \int 2 \cos X \cdot 2 \cos X dX$$

$$= \int 4 \cos^2 X dX = \int 4 \cdot \frac{1}{2} (1 + \cos 2X) dX$$

$$= \int 2(1 + \cos 2X) dX$$

$$= 2(X + \frac{1}{2} \sin 2X) + C$$

$$= 2 \arcsin(\frac{t}{2}) + \sin(2 \arcsin(\frac{t}{2})) + C$$

(b) $\int \frac{x+3}{x^2-1} dx$

$$\begin{aligned} \frac{x+3}{x^2-1} &= \frac{x+3}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1} \\ &= \frac{A(x-1) + B(x+1)}{(x+1)(x-1)} \\ &= \frac{(A+B)x - A + B}{(x+1)(x-1)} \end{aligned}$$

$$\begin{aligned} \Rightarrow A+B &= 1 \\ -A+B &= 3 \end{aligned} \Rightarrow A = -1, B = 2$$

$$\begin{aligned} \Rightarrow \int \frac{x+3}{x^2-1} dx &= \int \left(\frac{-1}{x+1} + \frac{2}{x-1} \right) dx \\ &= -\ln|x+1| + 2\ln|x-1| + C \end{aligned}$$

Problem 4. (18 points) Evaluate each of the following improper integrals or show that it is divergent.

(a) $\int_1^{+\infty} \frac{dy}{y(\ln y)^2}$

$$\int_1^{+\infty} \frac{dy}{y(\ln y)^2} = \int_1^2 \frac{dy}{y(\ln y)^2} + \int_2^{+\infty} \frac{dy}{y(\ln y)^2}$$

$$\begin{aligned} \text{check } \int_1^2 \frac{dy}{y(\ln y)^2} &= \lim_{t \rightarrow 1^+} \int_t^2 \frac{dy}{y(\ln y)^2} \\ &= \lim_{t \rightarrow 1^+} \left(-\frac{1}{\ln y} \right) \Big|_t^2 \\ &= \lim_{t \rightarrow 1^+} \left(-\frac{1}{\ln 2} + \frac{1}{\ln t} \right) = +\infty \\ &\Rightarrow \text{Divergent.} \end{aligned}$$

(b) $\int_2^{10} \frac{dx}{\sqrt[3]{10-x}}$

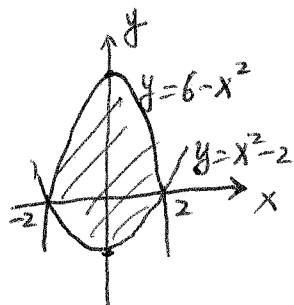
$$\begin{aligned} &= \lim_{t \rightarrow 10^-} \int_2^t \frac{dx}{\sqrt[3]{10-x}} \\ &= \lim_{t \rightarrow 10^-} \left(-\frac{3}{2} (10-x)^{2/3} \right) \Big|_2^t \\ &= \lim_{t \rightarrow 10^-} \left(-\frac{3}{2} (10-t)^{2/3} + \frac{3}{2} (10-2)^{2/3} \right) \\ &= 0 + \frac{3}{2} \cdot 4 \\ &= 6 \\ &\Rightarrow \text{convergent.} \end{aligned}$$

Problem 5. (24 points)

(a) Sketch the region enclosed by the curves $y = 6 - x^2$, $y = x^2 - 2$ and find its area.

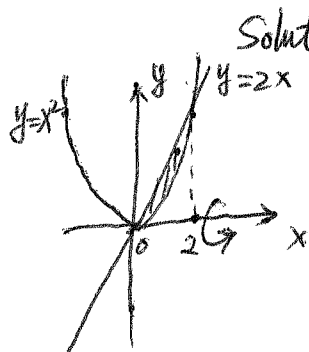
Solution:
$$\begin{cases} y = 6 - x^2 \\ y = x^2 - 2 \end{cases} \Rightarrow 6 - x^2 = x^2 - 2$$

$$2x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$$



$$\begin{aligned} A &= \int_{-2}^2 [(6 - x^2) - (x^2 - 2)] dx \\ &= \int_{-2}^2 [8 - 2x^2] dx \\ &= \left(8x - \frac{2}{3}x^3 \right) \Big|_{-2}^2 \\ &= \left(8 \cdot 2 - \frac{2}{3}2^3 \right) - \left(8 \cdot (-2) - \frac{2}{3}(-2)^3 \right) \\ &= \frac{32}{3} + \frac{32}{3} = \frac{64}{3} \end{aligned}$$

(b) Find the volume of the solid obtained by rotating the region bounded by the curves $y = 2x$, $y = x^2$ about the x -axis.



Solution:

$$\begin{cases} y = 2x \\ y = x^2 \end{cases} \Rightarrow x^2 = 2x$$

$$x(x - 2) = 0$$

$$x = 0, x = 2$$



$$\begin{aligned} A(x) &= \pi (2x)^2 - (x^2)^2 \\ &= \pi (4x^2 - x^4) \end{aligned}$$

$$\begin{aligned} V &= \int_0^2 A(x) dx \\ &= \int_0^2 \pi (4x^2 - x^4) dx \\ &= \pi \left(\frac{4x^3}{3} - \frac{x^5}{5} \right) \Big|_0^2 \\ &= \pi \left(\frac{4 \cdot 2^3}{3} - \frac{2^5}{5} \right) \\ &= \frac{64}{15} \pi \end{aligned}$$